Kiernan Preve

San Luis, Argentina · English C1 (TOEFL 106)
EU Citizenship (Italian)
kiernanpreve@gmail.com · github.com/KierPrev · kier.ar

Professional Summary

Physics student with an interdisciplinary background focused on the hardware-software bridge applied to biological systems. Specialist in the data value chain: from experimental design and capture via sensors and **Computer Vision (OpenCV)** to statistical analysis and modeling in Python/R. Active experience in the AgTech sector through magnetopriming research and the development of low-cost scientific instrumentation. Complementary background in bilingual science communication.

Key Skills

Scientific Computing: Python (pandas, NumPy, **OpenCV**, SciPy), R, SQL, Julia, Physical Modeling.
Hardware & IoT: Arduino (C/C++), Data Acquisition (DAQ) protocols, environmental sensors, open-source scientific hardware.
Methodologies & Ops: Git/GitHub, Linux (Fedora/Silverblue), Agile (Scrum), bilingual technical communication.
Languages: English (C1 - TOEFL 106), Spanish (Native), German (A1), French (A1).

Experience

Undergraduate Research Assistant (Thesis) — *MBLab (Magnetobiology Lab) — UNSL*
2025–Present

- Research on **Magnetopriming** in maize seeds: studying magnetic field effects on germination vigor and early growth.
- Developed a **Computer Vision system using OpenCV** for automated seed counting and radicle length measurement, significantly reducing human error and processing time.
- Performed experimental data analysis and growth curve modeling using Python.

Instrumentation Developer — *LibreLab UNSL*
2021–2022

- Prototyped low-cost sensors and scientific instrumentation based on open-source hardware.
- Integrated microcontrollers with visualization interfaces for real-time monitoring of physical variables.

Scrum Master (Internship) — *MakisanTech*
2024–Present

- Facilitated Agile workflows for a 5-person cross-functional team delivering a web-based appointment booking application.
- Coordinated alignment between hardware and software delivery streams across sprints.

Selected Projects

Automated Phenotyping (OpenCV): Image processing algorithm for morphological trait extraction in maize seedlings.
Low-cost IoT Rain Gauge: Arduino-based precipitation measurement system designed to improve local hydrological data resolution.

Education

Licenciatura en Física (Physics Degree — 5-year program) — Universidad Nacional de San Luis (UNSL)
2018–Present · Expected Graduation: Last Quarter 2026

Tecnicatura Universitaria en Producción Musical (University Technical Degree in Music Production) — Universidad Nacional de San Luis (UNSL)
2013–2017

Certifications & Courses

Python and Jupyter Notebook — UNSL (2018)
Java and SQL Programming — UNSL (2017)
Audio Mastering — Escuela Tecson (2016)

Science Communication

Podcast Host — *La curiosidad nos va a salvar*

- Independent science communication podcast translating physics and interdisciplinary research for general audiences.

NASA Space Apps Challenge — Space-themed narrative

- Wrote and narrated a science-outreach story for young audiences to build engagement with astrophysics, focused on solar flare phenomena.

Additional Information

Music Production: Arrangement, mixing, and mastering (Pro Tools, Studio One, Ableton, Cubase).
Writing: Published author of *Dentro del Origen* (ISBN 978-9878242514).